

# Sentinel AI

From Capstone Idea to a Working Prototype

Dataset Assessment, Data Preparation Plan, Transaction Flagging Policy,  
and Immediate Next Steps

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*This document extends the original Sentinel AI Idea & Implementation Guide with a concrete assessment of the three candidate datasets on hand, a data-preparation plan, a draft transaction-flagging policy (rules and parameters), and an ordered checklist of what to build next. Treat every threshold in this document as a starting point to be tuned against real pilot results, not a final answer.*

# 1. Where You Are Right Now

Mapped against the original 10-week plan (Section 6 of the Idea & Implementation Guide), you are entering **Phase 1: Data Pipeline & ML Baseline (Weeks 1-3)** — except you are starting with flat-file datasets instead of a live ERP connector, which actually removes the hardest and slowest part of Phase 1 (ERP integration). That means you can move straight to schema normalization and baseline modeling.

## What's already true:

- Three clean, labeled transaction populations are on hand (no nulls, no duplicate IDs).
- Each population has a plausible mapping to one of the verticals named in the original brief.
- Labels (fraud\_label) exist, which is a bonus - real audit engagements rarely have ground truth, so use these to build and validate your evaluation harness now, then treat unlabeled data as the real-world case.

## What's still missing before Phase 2 can start:

- No supporting-document text (invoice line items, wire purpose/memo, contract references) for the agent layer to retrieve - this blocks any real RAG/agent demo.
- No user/approver identity fields, so procurement-specific checks like duplicate-payment or maker-checker segregation-of-duties can only be partially simulated.
- No chart-of-accounts hierarchy - GL account IDs are flat codes with no category, which limits account-level baselining.

# 2. Dataset Assessment

All three files share an identical 14-column schema (transaction\_id, transaction\_time, amount, transaction\_type, customer\_id, merchant\_id, account\_id, location, device\_id, payment\_method, transaction\_frequency, balance\_before, balance\_after, fraud\_label). That uniformity is itself informative: it strongly suggests all three were produced by the same synthetic-data generator rather than sourced independently. That's fine for a prototype, but note it plainly if you present this as external validation data - don't imply it was pulled from three unrelated real-world sources.

| Dataset   | Transaction type(s)            | Rows    | Fraud rate | Avg / max amount   | Best-fit vertical                  |
|-----------|--------------------------------|---------|------------|--------------------|------------------------------------|
| corporate | Invoice Payment (1 type)       | 115,000 | 2.0%       | Rs 13.6K / Rs 150K | Procurement / AP duplicate-payment |
| aml       | International Wire (1 type)    | 115,000 | 1.5%       | Rs 8.2K / Rs 540K  | Banking-NBFC AML monitoring        |
| retail    | Transfer, Purchase, Withdrawal | 115,000 | 3.5%       | Rs 115 / Rs 8K     | Consumer/retail banking fraud      |

## 2.1 Should you merge them into one dataset?

**No - do not concatenate rows into a single training set.** The amount distributions differ by roughly two orders of magnitude between retail and corporate/AML. A single isolation forest or autoencoder trained on the pooled data will learn a distribution dominated by whichever population has the most rows at the wrong scale, and will

misjudge the other two - either flagging every retail transaction as anomalous or missing genuine corporate outliers entirely. This also directly matches guidance already in the original brief: *models can be retrained per client/industry rather than forcing one global model*.

### What to do instead:

- **Normalize, don't merge.** Map all three onto the common Layer-1 schema from the architecture (date, amount, account, counterparty, document reference) so the same ingestion code can read any of them.
- **Tag, don't blend.** Add a single *vertical* column (corporate / aml / retail) so a shared dashboard can display all three side by side without their statistics being pooled.
- **Train per-population.** Fit a separate isolation forest + autoencoder pair for each of the three - this mirrors exactly how you'd onboard three real clients.
- **Fabricate the missing evidence layer.** Generate a synthetic supporting-document field per transaction (e.g. an invoice description for corporate, a wire purpose code for AML, a merchant category note for retail) so the RAG/agent layer in Phase 2 has something concrete to retrieve and cite. Label it clearly as synthetic evidence in any write-up.

## 3. Data Preparation Plan (Do This Before Modeling)

| Step | Action                                                                                                                                                                                                                         | Output                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1    | Normalize schema across all three files onto one canonical column set; keep the vertical tag.                                                                                                                                  | One loader function, three clean frames |
| 2    | Engineer per-account/customer baseline features: rolling mean & std of amount, days since last transaction, transaction velocity (count per day), balance-depletion ratio ((balance_before - balance_after) / balance_before). | Feature table per vertical              |
| 3    | Run first-digit and second-digit Benford's Law checks on the amount field per account/vendor cohort (needs a minimum sample size per cohort - roughly 30+ transactions, or skip the check for thin cohorts).                   | Digit-deviation score per transaction   |
| 4    | Time-based train/calibration split (e.g. first 9 months to fit the baseline, last 3 months held out) rather than a random split, since anomaly detection should reflect a real monitoring cadence.                             | Train / holdout sets                    |
| 5    | Fit isolation forest + autoencoder per vertical on the training window; score the holdout.                                                                                                                                     | Per-transaction risk score              |
| 6    | Evaluate against fraud_label on the holdout only: precision, recall, and false-positive rate at several score thresholds.                                                                                                      | Threshold vs. false-positive-rate curve |
| 7    | Generate the synthetic supporting-evidence table (see 2.1) and embed it - this unblocks Phase 2.                                                                                                                               | Vector-store-ready evidence set         |

## 4. Transaction Flagging Policy - Rules & Parameters (Draft)

This is a starting draft for the policy document you asked to author yourself - use it as a skeleton and adjust every number against your own pilot results per Section 6/Phase 3 of the original guide. A workable policy for Sentinel AI has three layers that combine into one composite score, matching the architecture's Layer 2 design (ML ensemble + Benford's Law + rules).

### 4.1 Statistical / ML layer

| Signal                       | Method                                 | Suggested starting parameter                                          | Flag condition                                     |
|------------------------------|----------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------|
| Multivariate outlier score   | Isolation forest                       | contamination = 0.02-0.04 (match each vertical's observed fraud rate) | Anomaly score in top 2-4% for that vertical        |
| Reconstruction error         | Autoencoder                            | Threshold = mean + 3 std of reconstruction error on training window   | Reconstruction error exceeds threshold             |
| Digit-distribution deviation | Benford's Law (first-digit chi-square) | p-value < 0.01, cohort size >= 30 transactions                        | Cohort's digit distribution deviates significantly |

### 4.2 Rule-based hard flags (deterministic, explainable, run alongside the ML layer)

| Rule                          | Parameter                                                                                                   | Rationale                                                                |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Just-under-threshold amounts  | Transaction within 5% below a known approval/reporting limit (e.g. just under a Rs 5L wire-review limit)    | Classic structuring / approval-avoidance pattern                         |
| Round-number bias             | Amount is a round number (ends in 000 or 00) AND is in the top 10% of that account's amount range           | Fabricated figures skew toward round numbers                             |
| Velocity spike                | transaction_frequency for an account exceeds 3x its own 30-day rolling average                              | Sudden burst suggests account takeover or channel stuffing               |
| New counterparty + high value | First transaction with a given merchant/beneficiary AND amount in the top 10% for that account              | Unvetted counterparty combined with size is higher risk                  |
| Balance depletion             | (balance_before - balance_after) / balance_before > 0.5 in a single transaction                             | Large single-transaction drawdown warrants review regardless of ML score |
| Off-hours / weekend timing    | transaction_time outside 08:00-20:00 local, or on a weekend, for an account with no prior off-hours history | Deviation from an account's own established pattern                      |
| Dormant-account reactivation  | First transaction after 90+ days of inactivity on that account_id                                           | Reactivated dormant accounts are a known fraud vector                    |

### 4.3 Composite scoring and review tiers

Combine the two layers into one 0-100 composite score per transaction (for example: 50% weight on the normalized ML anomaly score, 30% on Benford deviation, 20% on the count of rule-based hard flags triggered, capped and rescaled). Route by tier:

| Tier                 | Composite score | Action                                                | SLA             |
|----------------------|-----------------|-------------------------------------------------------|-----------------|
| Auto-clear           | 0-39            | No case file generated; logged only                   | N/A             |
| Standard review      | 40-69           | Case file drafted by agent, queued for auditor review | 5 business days |
| Priority review      | 70-89           | Case file drafted, queued with priority flag          | 2 business days |
| Immediate escalation | 90-100          | Case file drafted, auditor + engagement lead notified | Same day        |

*Governance notes for the policy document: (1) every threshold above must be re-tuned per vertical/client during the Phase 3 parallel-run pilot against the 15% false-positive-rate target in the original brief; (2) log every rule/score that contributed to a flag so the agent's explanation can cite the specific trigger, not just a final number; (3) the policy itself should be version-controlled - changing a threshold is a documented policy change, not a silent code edit.*

## 5. Immediate Next Steps (Ordered)

- **Write the schema-normalization loader** for the three CSVs onto one canonical column set with a vertical tag - this is the actual, safe form of "merging" the three datasets.
- **Engineer the rolling/velocity/depletion features** listed in Section 3, per vertical.
- **Run Benford's Law + isolation forest + autoencoder** separately per vertical on a time-based split.
- **Build the evaluation harness** against fraud\_label and produce a precision/recall/false-positive curve per vertical - this satisfies Phase 1's exit criteria from the original guide.
- **Draft the synthetic supporting-evidence table** (invoice text / wire memo / merchant note) so the Phase 2 agent layer has something real to retrieve and cite.
- **Finalize your flagging policy document** using Section 4 as the skeleton - fill in your own team's thresholds once the evaluation harness gives you real precision/recall numbers to tune against.
- **Only then** start the RAG/agent prompt work (Phase 2) - building the agent before the scoring layer is validated means you'd be explaining flags you can't yet trust.

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*Source basis: reconstructed and extended from the team's Sentinel AI Idea & Implementation Guide (Deloitte Capstone 2026, Team Stack Synapse) and direct inspection of pro\_dataset\_corporate.csv, pro\_dataset\_aml.csv, and pro\_dataset\_retail.csv. All thresholds and weights in Section 4 are starting points for the team to validate, not measured or approved values.*